Create a Bash script to monitor CPU usage.

- **If CPU > 80%, then send an email notification.**

```
[root@ip-172-31-27-147 ~]# mkdir Scripts
[root@ip-172-31-27-147 ~]# ls
Scripts
[root@ip-172-31-27-147 ~]# cd Scripts
[root@ip-172-31-27-147 Scripts]# nano cpu_monitor.bash
[root@ip-172-31-27-147 Scripts]# chmod 755 cpu_momnitor.bash
chmod: cannot access 'cpu_momnitor.bash': No such file or director
y
[root@ip-172-31-27-147 Scripts]# chmod 755 cpu_monitor.bash
[root@ip-172-31-27-147 Scripts]# ./cpu_monitor.bash
Current CPU usage: 0%
CPU usage is normal.
[root@ip-172-31-27-147 Scripts]# cat cpu_monitor.bash
#!/bin/bash

THRESHOLD=80
EMAIL="abc123@gmail.com"

CPU_IDLE=$(mpstat 1 1 | awk '/Average:/ && $2=="all" {print $NF}')

CPU_USAGE=$(awk -v idle="$CPU_IDLE" 'BEGIN {printf "%.0f", 100-idle}')

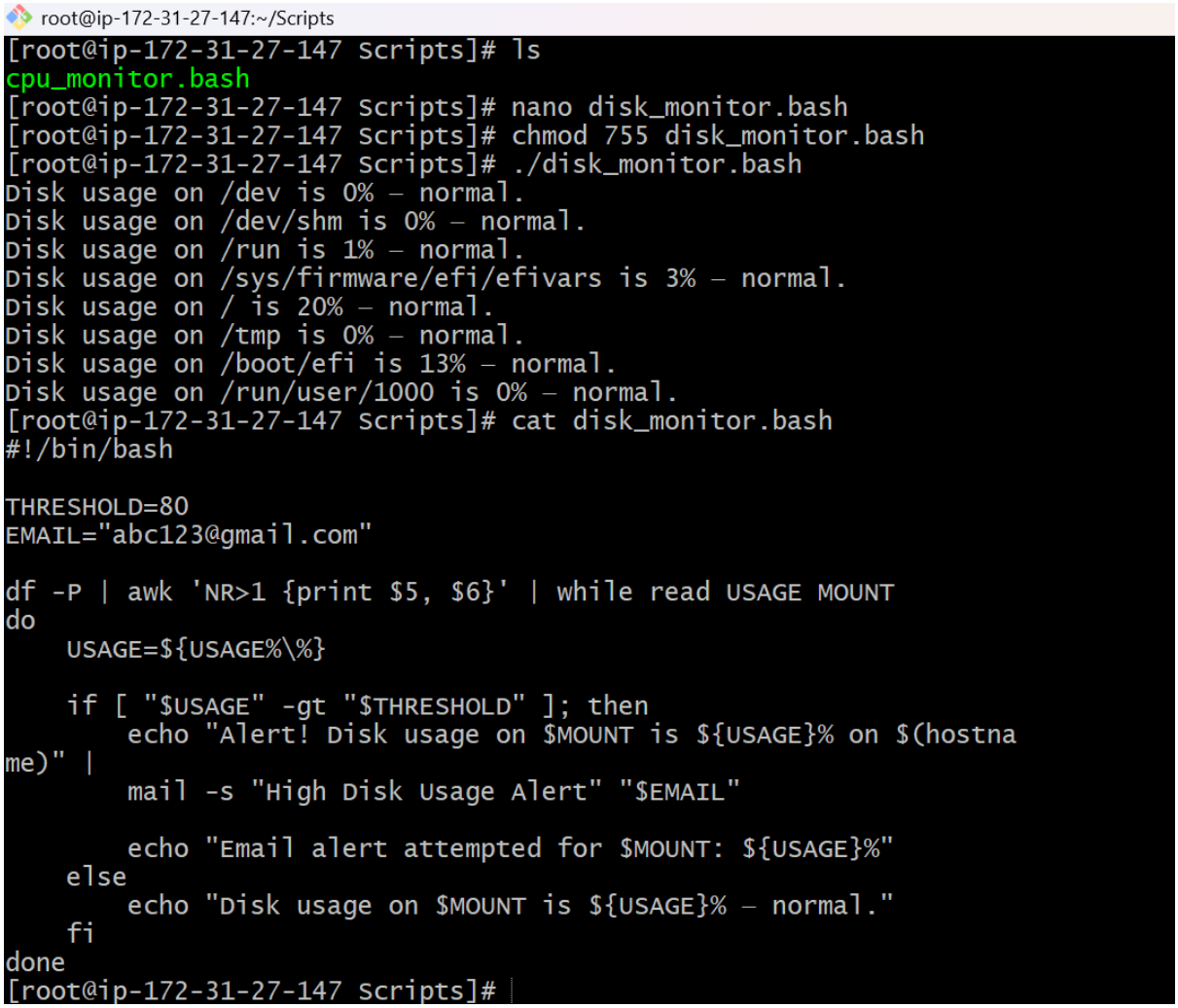
echo "Current CPU usage: ${CPU_USAGE}%"

if [ "$CPU_USAGE" -gt "$THRESHOLD" ]; then
    echo "Alert! CPU usage is ${CPU_USAGE}% on $(hostname)" |
    mail -s "High CPU Usage Alert" "$EMAIL"
    echo "Email alert sent."
else
    echo "CPU usage is normal."
fi
```

- Therefore, here we have created a directory script in order to store all our scripts based on bash.
- In that we have created a file naming cpu_monitor.bash to store the scripts to trigger automatic email notification if the cpu usage exceeds 80%.
- There we have created a script based on if condition.

Create a Bash script to monitor Disk Space.

- **If usage > 80%, then send an email notification.**



Disk Space Monitoring Using Bash Script

Objective

To monitor disk space usage on a Linux server and send an email notification if disk usage exceeds 80%.

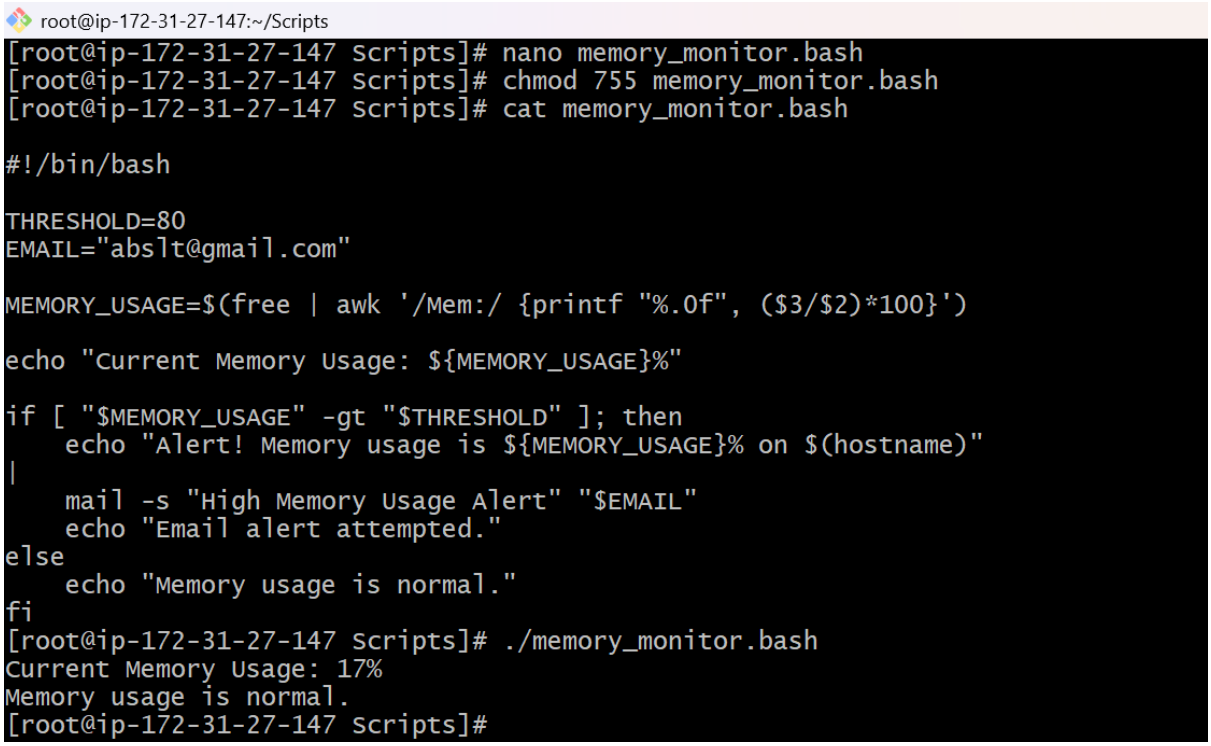
Explanation of the Script

1. #!/bin/bash
   Specifies that the script should run using the Bash shell.
2. THRESHOLD=80
   Sets the disk usage limit to 80%.
3. EMAIL="abc123@gmail.com"
   Specifies the email address that should receive the alert.

4. df -P
   Displays disk space usage for mounted filesystems in a consistent format.
5. awk 'NR>1 {print $5, $6}'
   Skips the heading and extracts the disk usage percentage and mount location.
6. while read USAGE MOUNT
   Reads each filesystem's usage percentage and mount location one by one.
7. USAGE=${USAGE%\%}
   Removes the % symbol so the usage can be compared as a number.
8. if ["$USAGE" -gt "$THRESHOLD"]
   Checks whether disk usage is greater than 80%.
9. mail -s "High Disk Usage Alert" "$EMAIL"
   Attempts to send an email notification if disk usage exceeds the limit. Email delivery requires a working mail configuration.
10. else
    Runs when disk usage is 80% or less and displays a normal status message.
11. done
    Marks the end of the loop that checks each filesystem.

Create a Bash script to monitor Memory Usage.

- **If usage > 80%, then send an email notification.**



1. Create a script: The nano memory_monitor.bash command creates or opens a file to write the Bash script.

2. Set the threshold: THRESHOLD=80 means the maximum allowed memory usage is 80%.

3. Set the email: The EMAIL variable stores the email address that should receive an alert.

4. Check memory usage: The free command gets memory information, and awk calculates the percentage of memory being used.

5. Display memory usage: The echo command displays the current memory usage on the screen.

6. Check the condition: If memory usage is greater than 80%, the script attempts to send an email notification. Otherwise, it displays that memory usage is normal.

7. Give execution permission: chmod 755 memory_monitor.bash allows the script owner to execute the file.

8. Run the script: ./memory_monitor.bash executes the script and displays the result.